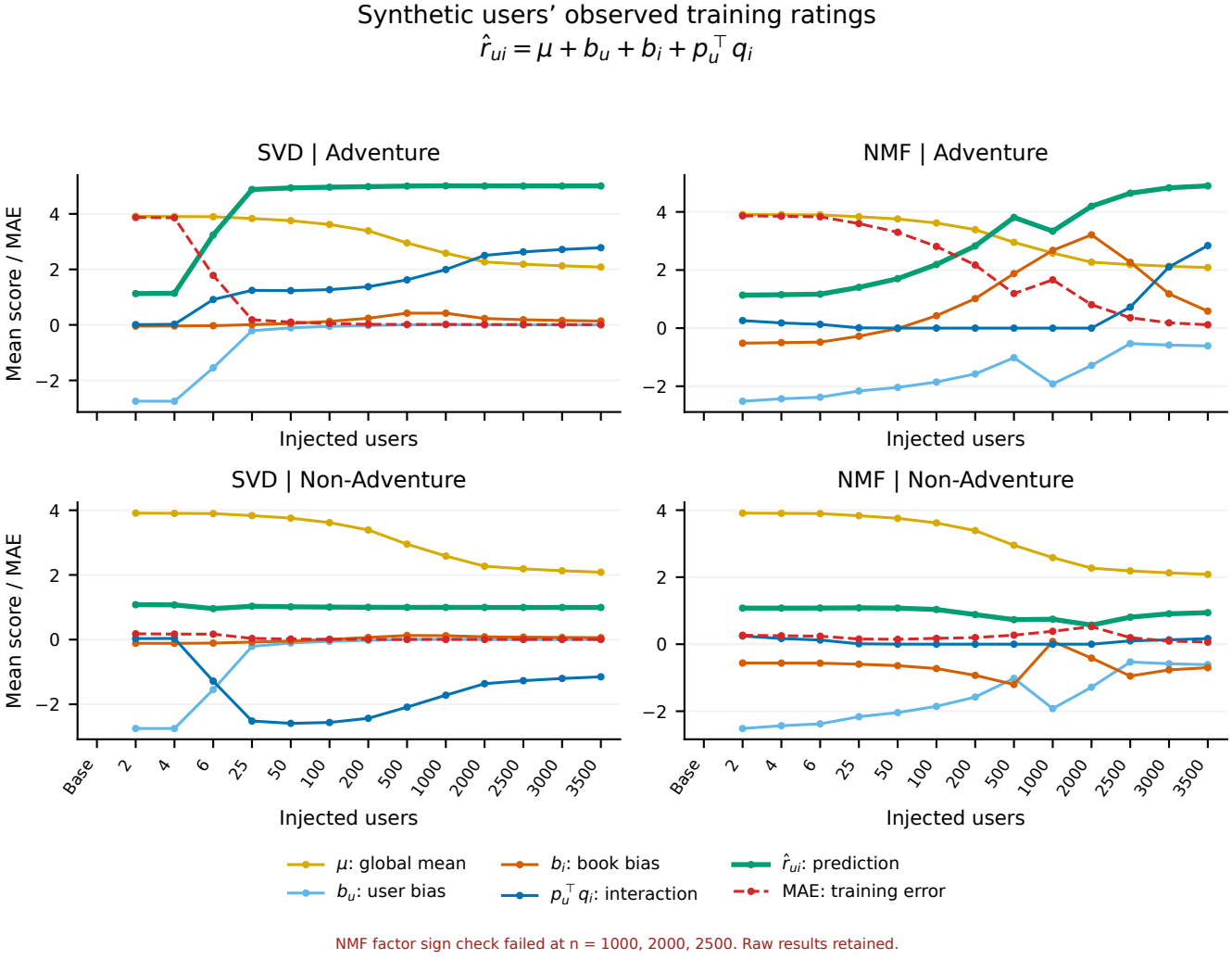
Synthetic users' observed training ratings
$\hat{r}_{ui} = \mu + b_u + b_i + p_u^\top q_i$
SVD | Adventure
NMF | Adventure
SVD | Non-Adventure
NMF | Non-Adventure
Mean score / MAE
Injected users
Base
2
4
6
25
50
100
200
500
1000
2000
2500
3000
3500
μ: global mean
b_u: user bias
b_i: book bias
$p_u^\top q_i$: interaction
$\hat{r}_{ui}$: prediction
MAE: training error
NMF factor sign check failed at n = 1000, 2000, 2500. Raw results retained.